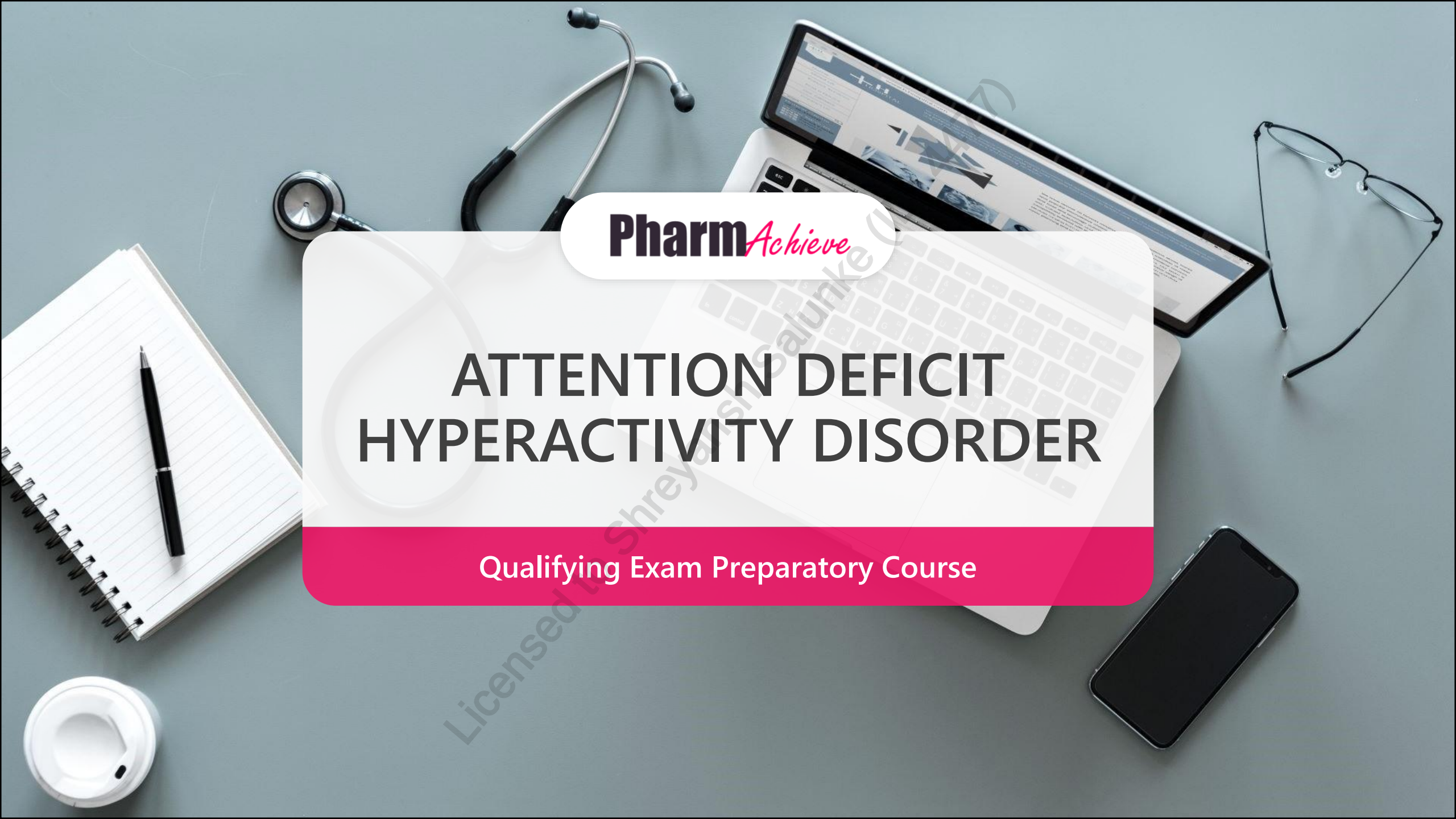


A top-down view of a grey desk with various items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill is faintly visible in the background.

Pharm*Achieve*

ATTENTION DEFICIT HYPERACTIVITY DISORDER

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ADHD Attention Deficit Hyperactivity Disorder

AE Adverse Event

AMP Amphetamine

BP Blood Pressure

CBT Cognitive Behavioural Therapy

CNS Central Nervous System

CV Cardiovascular

CYP Cytochrome P450

DR Delayed Release

DSM Diagnostic and Statistical Manual of Mental Disorders

ER/XR Extended Release

GI Gastrointestinal

HR Heart Rate

HS bedtime

IR Immediate Release

MPH Methylphenidate

MAOI Monoamine Oxidase Inhibitor

N/V Nausea/Vomiting

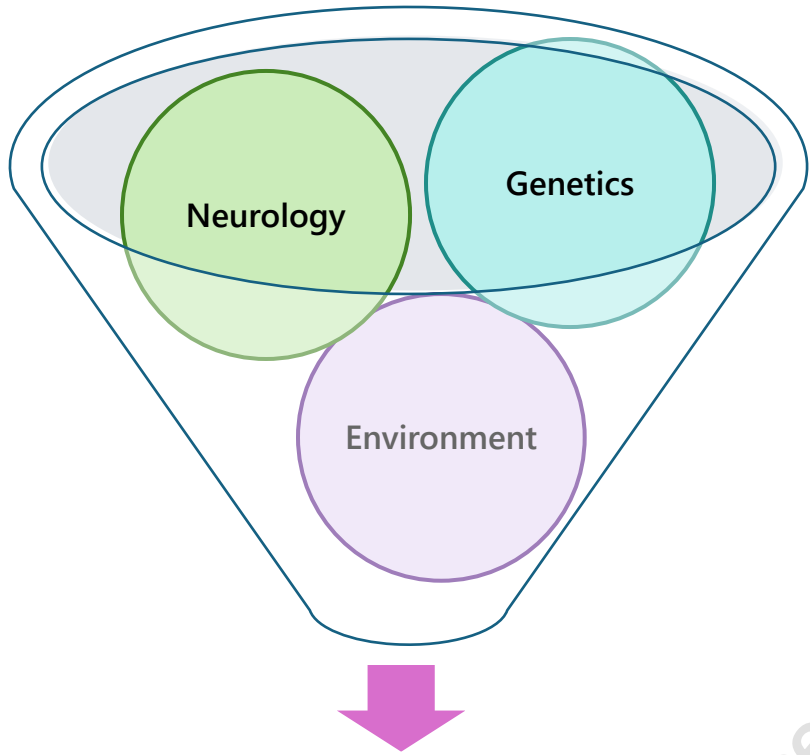
OROS Osmotic Release Oral System

SR Sustained Release

TCA Tricyclic Antidepressant

PATHOPHYSIOLOGY

FYI



Multiple gene variants contribute to ADHD risk

Genetic factors alter brain structure and function

Key regions affected: prefrontal cortex, basal ganglia, cerebellum

Catecholamine dysregulation

Neurochemical effects: **altered dopamine and norepinephrine** signaling

RISK FACTORS

Genetics

Premature birth/low birth weight

Neurological factors e.g. traumatic brain injury, epilepsy disorder

Exposure to environmental toxins (e.g. lead, pesticides)

Tobacco or alcohol use during pregnancy

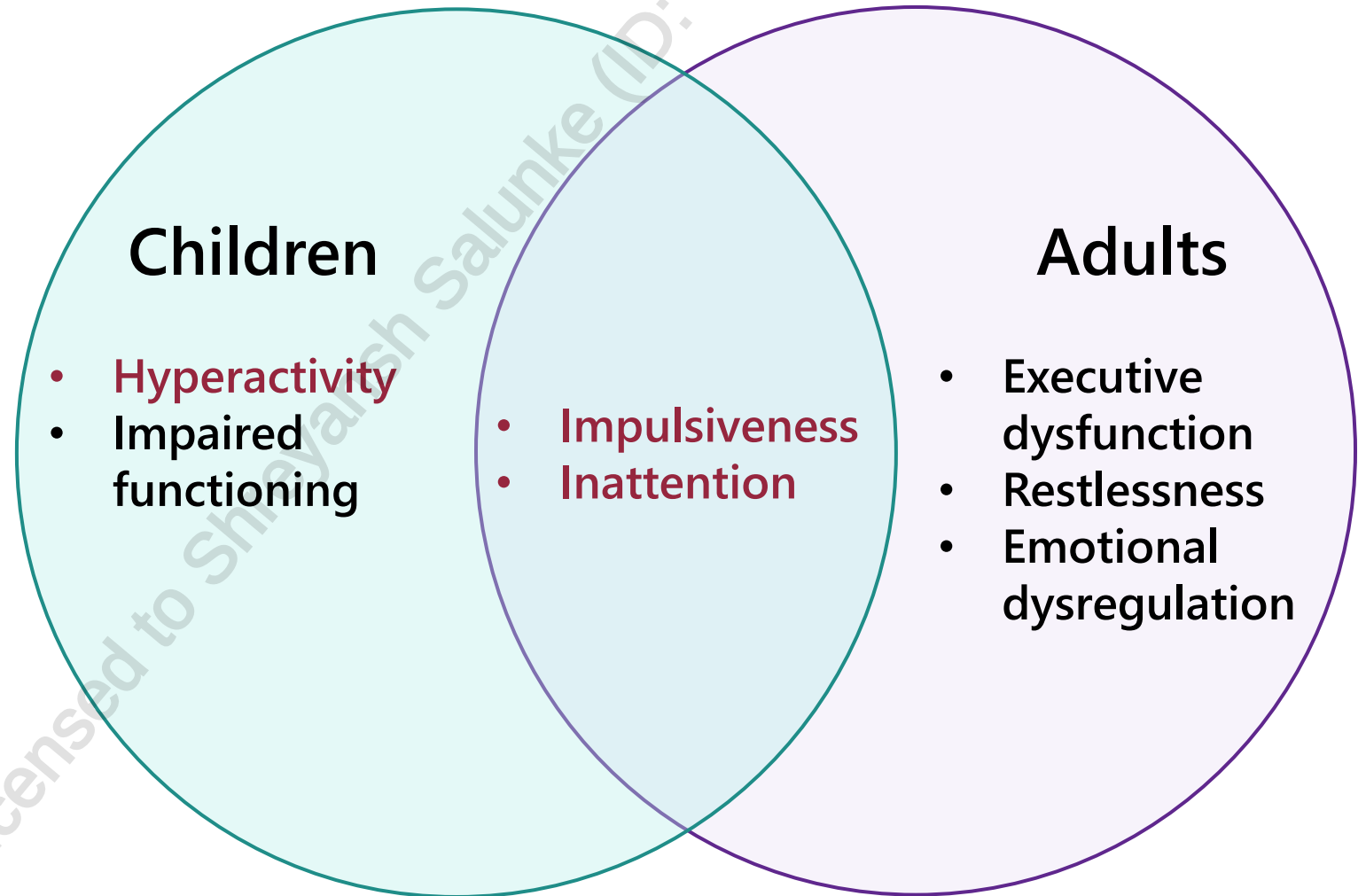
DSM-5 CRITERIA OF ADHD SYMPTOMS

Inattention Symptoms	Hyperactive-Impulsive Symptoms
• Often fails to give close attention to details or makes careless mistakes in schoolwork	• Often fidgets with hands or feet or squirms in seat
• Often has difficulty sustaining attention in tasks or play activities	• Often leaves seat in classroom when remaining seated is expected
• Often does not seem to listen when spoken to directly	• Often runs about or climbs excessively in situations where it is inappropriate
• Often does not follow through on instructions and fails to finish schoolwork	• Often has difficulty playing or engaging in leisure activities quietly
• Often has difficulty organizing tasks and activities	• Often is "on the go" or often acts as if "driven by a motor"
• Often avoids, dislikes, or reluctantly engages in tasks requiring sustained mental effort	• Often talks excessively
• Often loses things necessary for activities (e.g. school assignments, pencils, or books)	• Often blurts out answers to questions before the questions have been completed
• Often is distracted by extraneous stimuli	• Often has difficulty awaiting turn
• Often is forgetful in daily activities	• Often interrupts or intrudes on others (e.g. butts into conversations/games)

CLINICAL PRESENTATION

Hallmark symptoms

- ❑ Inattention
- ❑ Impulsivity
- ❑ Hyperactivity*



DIFFERENTIAL DIAGNOSIS

Conditions that may overlap with symptoms of ADHD

- Vision or hearing impairment
- Thyroid dysfunction
- Hypoglycemia
- Severe anemia
- Sleeping disorders
- Mood disorders e.g. depression, anxiety
- Substance use disorders
- Obsessive Compulsive Disorders
- Learning disabilities and behavioural disorders

Medications with psychomotor side effects that may mimic symptoms of ADHD

- Mood stabilizers
- Decongestants
- Beta-agonists

GOALS OF THERAPY

- ✓ Decrease/eliminate core ADHD symptoms
- ✓ Improve academic, behavioural and occupational performance
- ✓ Minimize adverse effects to medications
- ✓ Improve social functioning and self-esteem
- ✓ Improve quality of life
- ✓ Prevent or reduce the impact of comorbidities (e.g. anxiety, depression, substance use disorder)

NON-PHARMACOLOGICAL MEASURES



- ❑ 1st line in children < 6 years old
- ❑ Less effective than stimulants



Psychotherapy

- First line therapy for preschoolers
- Evidence for cognitive behavioural therapy in adults



1. Psychoeducation

- Educating the patient and family about ADHD (symptoms, treatment, comorbidities, impairment)



2. Psychosocial interventions

- a) Cognitive (e.g. mindfulness-based therapy, neurofeedback) or
- b) Behavioural management (e.g. sleep hygiene, daily routine)



Physical Exercise

- Aerobic exercise and yoga may to improve symptoms and cognitive function

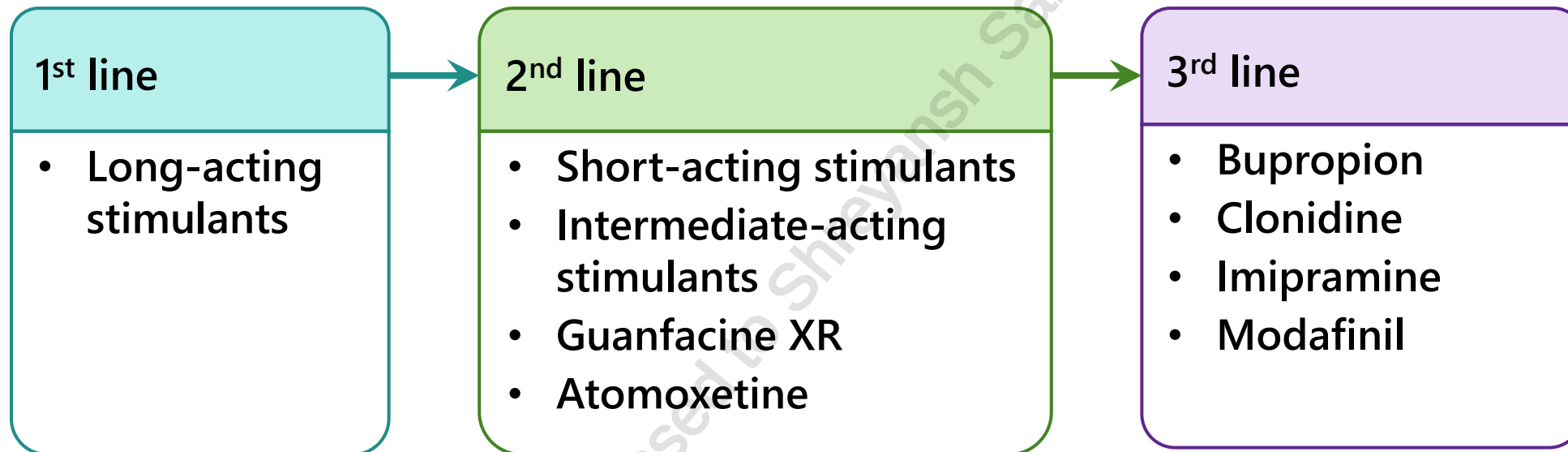


Education Considerations

- Classroom management (e.g. clear and precise directions)
- Organization skills training (e.g. timers/ apps for reminders)
- Behavioural peer interventions

PHARMACOLOGICAL TREATMENT

- Stimulants have a faster onset of action (1-3 weeks) compared with non-stimulants (1-6 weeks and 4-12 weeks for maximum effect)
- Adequate trial of long-acting stimulants is warranted before moving on to second-line therapy
- Stimulants: Amphetamine (AMP) based products and Methylphenidate (MPH) based products



SAFETY CONSIDERATIONS WITH STIMULANTS

Precautions

- Epilepsy
- Anxiety
- Tic disorders
- Renal impairment
- Peripheral vasculopathy (including Raynaud's phenomenon)
- History of substance abuse

Contraindications

- History of psychosis or mania
- Atherosclerosis, uncontrolled BP
- Uncontrolled hyperthyroidism
- Pheochromocytoma
- Narrow-angle glaucoma
- MAOI use within 14 days

TYPES OF STIMULANTS

Long-acting stimulants

- ☐ Better risk-benefit profile and evidence for effectiveness
- ☐ Sustained release formulations help maintain privacy
- ☐ Reduces risk of diversion/abuse and rebound hyperactivity
- ☐ Better tolerability

Short-acting stimulants

- ☐ Selected in patients who experience side effects, reduced tolerability, sub-optimal response with first-line therapy
- ☐ Can be used in combination therapy with first-line agents to **augment suboptimal responders or evening doses**

AMPHETAMINE-BASED STIMULANTS

*First Line
*Second Line

Amphetamine-based stimulants: Increase dopamine and norepinephrine.
Onset: 1-3 weeks, indicated in children ≥ 6 years and adults

Medication	Delivery	Duration of Action	Starting Dose	IR/DR %	Dose Titration
Adderall XR® Amphetamine Mixed Salts (capsules)	Beads can be sprinkled on soft foods Do not crush/chew beads	~12 h	5-10 mg daily QAM	50/50	↑5-10 mg at weekly intervals. Max dose/day: 20-30mg (all ages)
Vyvanse® Lisdexamfetamine	Capsule can be opened and contents diluted in liquid or sprinkled. Chewable tablets should be chewed thoroughly	~13-14 h	20-30 mg daily QAM	Continuous (prodrug)	↑10-20 mg by clinical discretion at weekly intervals. Max dose/day: 60mg (all ages)
Dexedrine® Dextroamphetamine (tablets and spansules)	Scored tablet can be cut in two, do not crush Spansules swallowed whole or can be opened and contents sprinkled on soft foods	~4 h (tablets) ~6-8 h (spansules)	Tab: 2.5-5 mg BID Span: 10 mg daily QAM	Tab: 100/0 Span: 50/50	↑5 mg at weekly intervals. Max dose/day: 40 mg (all ages)

UNIQUE AMP-BASED STIMULANT DRUG INTERACTIONS

Drug classes	Example of agent	Interaction & recommendation
Acidifying agents ¹	Orange juice, vitamin C (ascorbic acid)	• ↓ amphetamine absorption, plasma levels, and ↑ excretion • Monitor side effects
Alkalinizing agents ¹	Sodium bicarbonate	• ↑ amphetamine absorption, plasma levels, and ↓ excretion, increase side effects • Monitor side effects or switch therapy

¹Vyvanse® (lisdexamfetamine) is a **prodrug**, it is not prone to pH-mediated food or drug interactions

MPH-BASED STIMULANTS

MOA: Increase Dopamine and Norepinephrine.
Onset: 1-3 weeks, indicated in children ≥ 6 years

*First Line
*Second Line

Methylphenidate	Delivery	Duration of Action	Starting Dose	IR/DR %	Dose Titration
Biphentin® (capsules)	Granules can be sprinkled, (do not crush/chew beads)	~10-12 h	10-20 mg daily QAM	40/60	↑ 10mg at weekly intervals. Max dose/day: 60 mg (children/adolescents) and 80 mg (adults)
Concerta® (ER tablets)	Osmotic-Controlled Release Oral Delivery System (OROS®), do not crush	~12 h	18 mg daily QAM	22/78	↑ 18mg at weekly intervals. Max dose/day: 54 mg (children/adolescents) and 72 mg (adults)
Foquest® (capsules)	Granules can be sprinkled, (do not crush/chew beads)	~16 h	25 mg daily QAM	20/80	↑ 10-15 mg in 5-day intervals. Max dose/day: 70 mg (children/adolescents) and 100 mg (adults)
Quillivant® ER (solution or chewable tabs)	Chewable tablets Powder for oral solution	~8-12 h ~12 h	20 mg daily QAM	30/70 20/80	Titrate up and down weekly by increments of 10mg Max dose/day = 60 mg
Methylphenidate Short-acting (tablets)	Scored tablets can be cut in two IR formulations can be crushed, but not recommended due to increased risk of abuse	~3-4 h	5 mg BID-TID	100/0	↑ 5-10 mg at weekly intervals. Max dose/day: 60 mg (all ages)

FYI – JORNAY PM®: the newest approved MPH-based stimulant. Taken once daily in the evening.

SIDE EFFECTS OF STIMULANTS

Common side effects

- Insomnia
- Headache
- N/V, ↓ appetite, dry mouth
- Slowed growth: ↓ weight and height
 - **Get height/weight at baseline and monitor every 3-6 months**
- Nasal congestion, nasopharyngitis (methylphenidate)
- ↑ BP/HR
 - **Monitor prior to initiation and during follow-up**

Other side effects

- Sexual dysfunction
- Hallucinations, psychosis, aggression
- Suicidal thoughts
- Uncovering tics
- **Decreased seizure threshold**
- Sudden death associated with pre-existing CV disease
- Blood dyscrasias
- Priapism (atomoxetine > methylphenidate)
- Rhabdomyolysis

MANAGEMENT OF SIDE EFFECTS ASSOCIATED WITH STIMULANTS

Anorexia	• Give high calorie meal when drug effect minimal (AM and HS)
Stomach upset	• Give after eating (full stomach)
Insomnia	• Give dose earlier in the day • Change to shorter-acting formulation
Headache	• Give in divided dose with food; may require analgesic
Decreased growth	• Consider drug holiday, use atomoxetine over stimulant
Irritability, aggression, mania	• ↓ dose or discontinue, refer for reassessment
Tics/abnormal movement	• ↓ dose; consider using atomoxetine or guanfacine instead
↑ HR/↑ BP	• ↓ dose or change formulation; switch to atomoxetine
Hallucination	• Discontinue stimulant, refer for reassessment

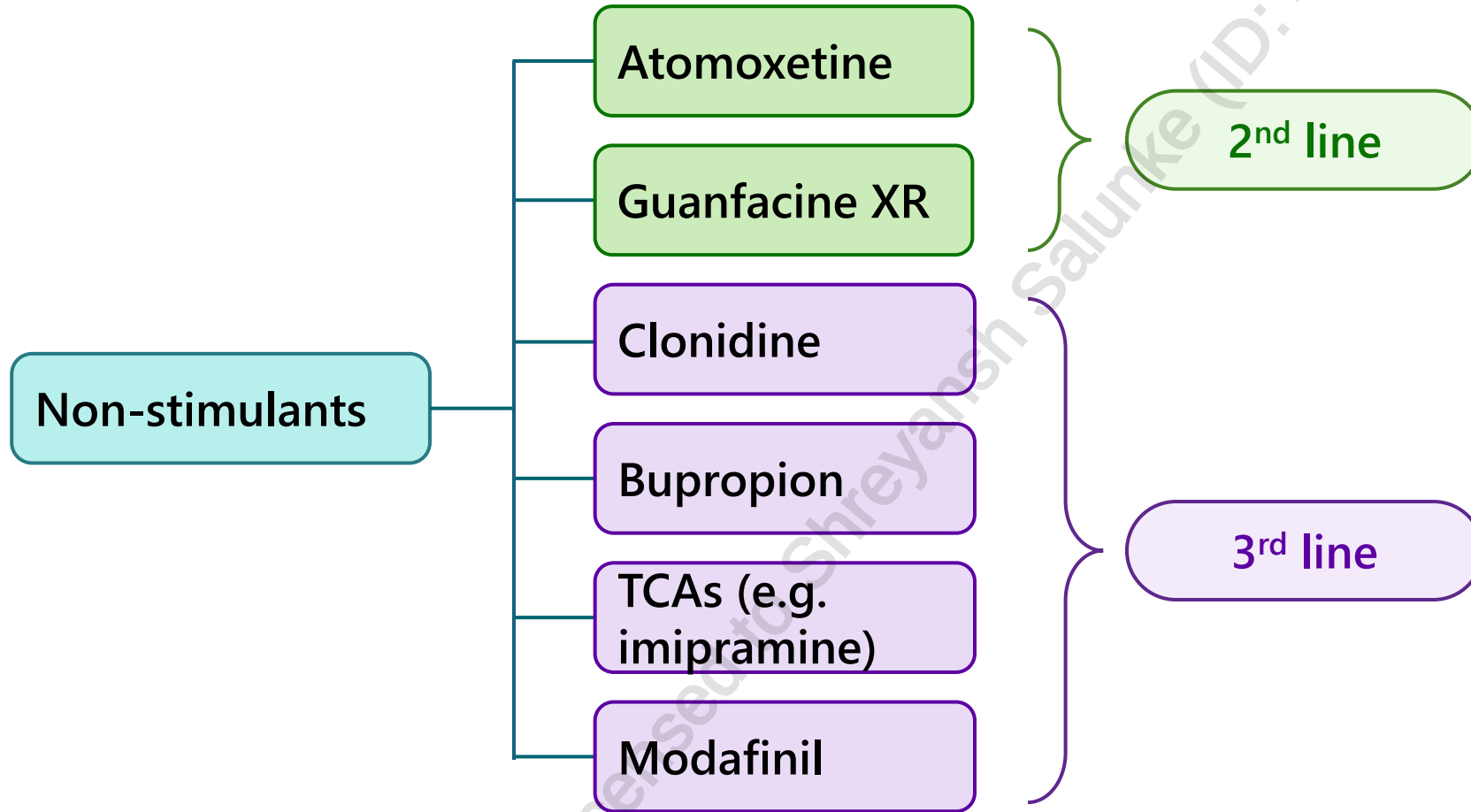
DRUG HOLIDAYS

Definition	Suggested implication (benefits outweigh risks)	Recommended against (risk outweigh benefits)
• Planned, temporary break from daily medication use • Done on a case-by-case basis if benefits > risks • Usually done during summer break for children	• Discontinue medication to re-evaluate ADHD symptoms (case-by-case basis) • CADDRA: While drug holidays can be considered for appetite and growth side effects, dose reduction or switching to a different medication are also important options	• Canadian Pediatric Society: drug holidays are not routinely recommended, unless there are specific concerns about a medication's adverse effects • Continuous drug holidays can increase psychostimulant side effects

If the decision to taper is made by the physician, and the patient/ family member considers:

- ☐ Taper stimulants to avoid withdrawal
- ☐ Atomoxetine is less likely to produce withdrawal
- ☐ Taper guanfacine XR and clonidine to reduce the risk of rapid BP increase

NON-STIMULANT DRUG THERAPY OPTIONS



ATOMOXETINE (NOREPINEPHRINE REUPTAKE INHIBITOR)

Medication	Delivery	Duration of Action	Starting Dose	Dose Titration
Atomoxetine capsules	Capsule needs to be swallowed whole to reduce GI side effects	Up to 24 h	Children 6+: 0.5 mg/kg/day Adults: 40 mg daily for 7-14 days	Maintain dose for a minimum of 7-14 days before adjusting: Children: 0.8 then 1.2 mg/kg/day Adults or children >70 kg: 60 then 80 mg/day Max dose/day: 1.4 mg/kg/day or 100 mg

- Mechanism of action: NE reuptake inhibitor, not a CNS stimulant (no known abuse potential)
- Drug Interactions: Metabolized by CYP2D6 (genetic polymorphism caution) – bupropion, fluoxetine, paroxetine, etc.
- Onset of benefit: up to 6-8 weeks for maximal benefit
- Role in therapy: second-line if patient does not respond well/tolerate adequate trial of stimulants or has comorbid substance abuse disorder/depression, if comorbid Tourette's syndrome or tics disorder - *Less effective than stimulants*

Side effects

- N/V, decreased appetite, weight loss, cough, mild increase in BP and HR
- Rare/ severe: seizures, suicidal thoughts, hepatic toxicity, priapism, increased QT
- Less growth suppression than stimulants

Monitor during treatment

- Priapism
- Signs of liver injury
- Delayed growth

Contraindications:

- MAOI within 14 days
- Symptomatic CVD, atherosclerosis, moderate to severe hypertension,
- Uncontrolled hyperthyroidism, pheochromocytoma,
- Narrow-angle glaucoma

GUANFACINE (ALPHA-2A AGONIST)

*First Line
*Second Line

Medication	Delivery	Duration of Action	Starting Dose	Dose Titration
Intuniv XR® (Guanfacine XR tablets)	Pills need to be swallowed whole to keep delivery mechanism intact (do not administer with high fat meals, avoid grapefruit juice)	Up to 24 h	Children 6-17 years: 1 mg daily (AM or PM) Not indicated in adults	Maintain dose for a minimum of 7 days before adjusting by ≤ 1 mg weekly Max dose/day: Monotherapy: 6-12 years: 4 mg 13-17 years: 7 mg As adjunctive therapy: 6-17 years: 4 mg

- **Mechanism of Action:** Selective Alpha-2A agonist, **NOT a CNS stimulant** – (no known abuse potential)
- **Drug Interactions:** Metabolized by CYP3A4 – azole antifungals, diltiazem, etc.
- **Onset of Benefit:** up to 4 weeks for maximal benefit
- **Role in therapy:** 2nd line if patient does not respond well/tolerate adequate trial of stimulants, if comorbid substance abuse disorder/depression/ Tourette's syndrome/ tics disorder/ oppositional defiant disorder or anxiety disorder. Reduces symptoms of aggression, impulsivity and hyperactivity (less benefits for inattention) - **Less effective than stimulants**
- **Common side effects:** sedation and somnolence, decreased HR and BP, fatigue, constipation, headaches
- **If discontinuing, must be slowly tapered to avoid hypertensive crisis**

THIRD-LINE PHARMACOLOGICAL AGENTS

FYI

- Less effective than stimulant medication, reserved for treatment-resistant cases
- May benefit those patients with “comorbid” conditions or as adjunctive therapy:
 - Bupropion, TCAs (*e.g. imipramine*), Clonidine, Modafinil
 - Choice of agent is based on SE profile (risk vs. benefit)
 - Bupropion: useful for comorbid depression
 - Clonidine: helpful with tics, hyperactivity and impulsivity, less effective for inattention

PREGNANCY AND BREASTFEEDING

- ❑ **First line** → non-pharmacologic measures
- ❑ Medications may be considered on a **risk/benefit basis**

Pregnancy

Severity	Recommendation
Mild to moderate	Non-pharmacological (e.g. psychoeducation, coaching, CBT, mindfulness-based interventions)
Moderate to severe	- Non-pharmacological + pharmacological - AMP or MPH stimulants - Non-stimulants: use with caution 

Breastfeeding

- Pharmacotherapy can be considered using a risk vs. benefit analysis and a shared decision-making process. If taking:
 - Amphetamines**: transfers into breast milk, but use appears to be safe
 - Effects on neurological development are not well studied
 - Methylphenidate**: Low transfer into breastmilk, effects on neurological development are not well studied
 - If using any stimulants, monitor breastfed infants for insomnia, agitation, irritability and poor weight gain
 - If using **bupropion**, monitor breastfed infants for sedation, weight gain and irritability
 - Clonidine and guanfacine XR** may decrease milk production, other options preferred
- 

MONITORING PARAMETERS

Efficacy Endpoints		
Parameter	Degree of Change	Timeframe
Behaviour (Academics, Sleep, Appetite)	Improvement	Regularly, medication dependent
Vitals and growth suppression (BP, HR, Weight and Height)	Normal	Baseline, then at each visit to doctor
Risk of substance abuse	None	Ongoing
Safety Endpoints		
Medication AEs	Prevent or minimize	Ongoing

TAKEAWAY

- **First-line medications**

- Children < 6 → non-pharmacologic measures
- Children/adults ≥ 6 → long-acting stimulants + non-pharmacologic measures
- Patients should try long-acting amphetamine-based and methylphenidate-based stimulants prior to moving to 2nd line therapies
- Stimulant medications may slow growth and increase HR and BP

- **Second-line medications**

- Short/intermediate-acting stimulants, atomoxetine, and guanfacine
 - Guanfacine only has indication for children 6-17
- Non-stimulant medications are not as effective as stimulant medications

- **Special populations**

- During pregnancy and breastfeeding, non-pharmacologic measures are the preferred first-line therapy.
 - Pharmacotherapy may be considered on a case-by-case basis, carefully weighing the potential benefits against the risks

CASE 1

BB is a 13-year-old girl who presents to your primary care clinic accompanied by her mother. During the physician's assessment, BB doesn't seem to be listening and is very easily distracted. However, she remains in her seat for the entire appointment. BB is a Grade 8 student (attends class from 08:30-15:00). For the past 8 months, BB's teacher reports that she is not performing well on her assignments and tests. She seems unfocused, her answers often lack detail, and sometimes entire sections are left blank.

BB has no significant medical history and does not take any medications or vitamins. BB's blood pressure is 103/69 mmHg, her HR is 88 bpm, and she weighs 45 kg. BB is not yet able to swallow tablets or capsules.

1. What is the likely diagnosis for BB?
 - a) ADHD, predominately inattentive
 - b) ADHD, predominately hyperactive/impulsive
 - c) ADHD, combined inattentive and hyperactive/impulsive
 - d) Oppositional Defiant Disorder (ODD)



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BB has no significant medical history and does not take any medications or vitamins. BB's blood pressure is 103/69 mmHg, her HR is 88 bpm, and she weighs 45 kg. BB is not yet able to swallow tablets or capsules.

2. What is the first treatment strategy you would recommend for BB?
- a) Nonpharmacological therapy alone
 - b) Stimulant therapy + nonpharmacological therapy
 - c) Alpha-adrenergic therapy + nonpharmacological therapy
 - d) Stimulant therapy alone



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BB has no significant medical history and does not take any medications or vitamins. BB's blood pressure is 103/69 mmHg, her HR is 88 bpm, and she weighs 45 kg. BB is not yet able to swallow tablets or capsules.

3. Which of the following stimulants would be most appropriate for BB?
- a) Methylphenidate
 - b) Adderall XR®
 - c) Concerta ER®
 - d) Dexedrine®

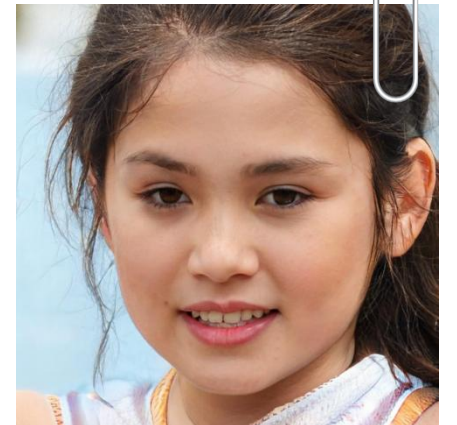


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BB has no significant medical history and does not take any medications or vitamins. BB's blood pressure is 103/69 mmHg, her HR is 88 bpm, and she weighs 45 kg. BB is not yet able to swallow tablets or capsules.

4. BB was prescribed Vyvanse® (lisdexamfetamine) chewable tablets. Which of the following adverse effects is she LEAST likely to experience?
- a) Insomnia
 - b) Tremor
 - c) Weight gain
 - d) Dry mouth



REFERENCES

1. American Journal of Obstetrics & Gynecology article S0002-9378(24)00376-4. *Am J Obstet Gynecol*. 2024;[Epub ahead of print]. doi:10.1016/j.ajog.2024.XXXX.
2. Canadian ADHD Resource Alliance. *Canadian ADHD Practice Guidelines 4.1*. CADDRA; January 6, 2021.
3. Bélanger SA, Andrews D, Gray C, Korczak D; Canadian Paediatric Society, Mental Health and Developmental Disabilities Committee. ADHD in children and youth: Part 1—Etiology, diagnosis, and comorbidity. *Paediatr Child Health*. 2018;23(7):447-453. Reaffirmed January 11, 2024.
4. Feldman ME, Charach A, Bélanger SA; Canadian Paediatric Society, Mental Health and Developmental Disabilities Committee. ADHD in children and youth: Part 2—Treatment. *Paediatr Child Health*. 2018;23(7):462-472. Reaffirmed January 11, 2024.
5. Canadian ADHD Resource Alliance. *ADHD Medication Chart (EN-FR)*. CADDRA; October 2024.
6. UpToDate. *Attention-deficit hyperactivity disorder in adults: Treatment overview*. UpToDate.
7. UpToDate. *Attention-deficit hyperactivity disorder in adults: Epidemiology, clinical features, assessment and diagnosis*. UpToDate.
8. UpToDate. *Attention-deficit hyperactivity disorder in children and adolescents: Clinical features and diagnosis*. UpToDate.

CHANGE LOG

June 2025

- Slide 10: Updated 1st line in children < 6 years old, parent behaviour training, and removed dietary changes
- Slide 27: question 3 option a changed

August 2025

- Significant changes made throughout

October 2025

- Slide 11: Updated list of contraindications
- Slide 13: Added that Dexedrine spansules can be opened and sprinkled

November 2025

- Formatting changes made throughout
- Slide 7: Added details to clinical presentation of ADHD
- Slide 8: Added more conditions that may overlap with ADHD symptoms
- Slide 16: Added that methylphenidate IR can be crushed but not recommended
- Slides 27-30: Minor changes to the case and associated questions

March 2026

- Slide 2: Updated NAPRA competencies
- Slide 16: Added JORNAY PM®
- Minor formatting changes made throughout